

TRB-015 — Discovery

Submitter: **Jason Freeman**

Request: *Submission attempt failed for Elite Home Solutions*

What is proposed

Three coordinated code fixes plus a one-off data correction, shipped together so the same class of failure cannot recur and Jason's pending submission can clear today.

****1. UI — restrict the "Owner Type" dropdown**** (`src/pages/CustomerForm.tsx:363-373`)

Remove "Accountant" and any other non-ownership values from the ****Owner Type****. The options must be limited to values that map cleanly to ATAP's Ownership Type codes: ``Owner``, ``Corporate Officer``, ``Partner``, ``Member``, ``Shareholder``, ``Trustee``. Add a helper caption directly below the label — ****"How is this person related to the business? (For who is filing on their behalf, use Contact's Role above.)"** — to disambiguate from the ****Contact's Role**** select directly above it (which legitimately keeps Accountant/Bookkeeper, since that field describes the filer, not the owner).

****2. Defensive mapper + fast-fail selector**** (`netlify/functions/lib/arkansas-form-automation.ts:1517`` and the ``#Dm-21`` fill site)

Introduce a ``VALID_OWNERSHIP_CODES = new Set(['OWN','CRP','PAR','MEM','SHR','TRU'])`` allow-list. After the existing lookup chain, validate the resolved code against the set; if it fails (e.g. ``ACT`` from ``CONTACT_TYPE_MAP['Accountant']`` at line 214), log a structured warning (``{customer_id, contact_type, resolved_code, offered_options}``) and fall back to ``OWN``. Before ``page.selectOption('#Dm-21', code)``, add ``await page.waitForSelector('#Dm-21 option[value=' + code + ''], { timeout: 5000 })`` so a missing option fails in 5s with the actual DOM options enumerated in the log, not after 30s of silent waiting.

****3. Per-field timeout isolation in Step 6**** (`arkansas-form-automation.ts``, Step 6 / Wizard Page 4 owner-info block)

Wrap each owner-info fill (First Name, Last Name, SSN, Phone, Ownership Type, Contact's Role) in its own ``try/catch`` that logs and continues rather than aborting the step. ATAP's own validator will then surface the real missing field, instead of the current misleading "Owner's First Name: Required" that fires because the prior ``selectOption`` timeout aborted every fill after it.

****4. Data correction — Elite Home Solutions**** (one-off, ships independent of code)

Run via Supabase MCP:

```
```sql
UPDATE customers
SET contact_type = 'Owner'
WHERE id = '086901c5-bc82-4620-99c9-a9226035e433';
```
```

This unblocks Jason's retry of submission ``36261872-faae-406c-bbac-864beb83103e`` (6 invoices, \$26,961.94) immediately; credit is already refunded (``payment_state = 'released'``).

****5. Vitest regression**** (`tests/arkansas-form-automation.ownership-mapper.test.ts`)

Cover three cases against the new mapper: (a) `contact\_type='Owner'` → `OWN`; (b)

`contact\_type='Accountant'` → `OWN` with warning logged (the TRB-015 case); (c) any unknown string → `OWN` with warning. Anchored on the actual exported mapper function so the test fails if the allow-list is bypassed.

****Out of scope:**** any other field on `CustomerForm.tsx` beyond the Owner Type dropdown; the Contact's Role dropdown above it is correct and stays unchanged.

Desired outcome

Jason re-submits Elite Home Solutions and the ATAP wizard advances through Step 6 to completion. No future customer can pick an Owner Type value that ATAP's `#Dm-21` does not accept, and if ATAP changes its allowed codes upstream, the automation fails in 5 seconds with a diagnostic log naming the offending code and the live options list — not in 30 seconds with a misleading "missing first name" symptom.

Success criteria

- `CustomerForm.tsx` Owner Type dropdown offers only ownership-category values; "Accountant" no longer appears as a selectable option in the rendered UI.
- Submitting a customer whose `contact\_type` is somehow set to `Accountant` (legacy data) results in ATAP receiving `OWN` and a warning row in the function logs — not a 30-second `selectOption` timeout.
- Elite Home Solutions row `086901c5-bc82-4620-99c9-a9226035e433` has `contact\_type = 'Owner'` after the SQL runs; Jason's next submission attempt clears Step 6 and reaches the confirmation page.
- New vitest suite passes in CI and fails if the allow-list guard is removed from `arkansas-form-automation.ts:1517`.
- Step 6 per-field try/catch verified by a manual run with an invalid Ownership Type: ATAP's own validator surfaces the genuine missing field rather than the cascaded "Owner's First Name: Required" red herring.